Two-dose vaccine effectiveness following the first reactive mass vaccination campaign against hepatitis E in Bentiu, South Sudan

Robin Nesbitt[1], John Rumunu[2], **Vincent Kinya Asilaza[3]**, Priscillah Gitahi[3], Patrick Nkemenang[3], Melat Haile[4], Jetske Duncker[3], Zelie Antier[3], Etienne Gignoux[4,1], Manuel Albela[4], Primitive Gakima[4], Joseph Wamala[5], Kediende Chong[2], Catia Alvarez[6], Isabella Eckerle[6], Monica Rull[4], Iza Ciglenecki[4] Andrew Azman[4,7].

[1]Epicentre, Paris, France; [2]Ministry of Health (MoH), Juba, South Sudan; [3]Médecins Sans Frontières (MSF), Juba, South Sudan; [4]MSF, Geneva, Switzerland; [5]WHO, Juba, South Sudan; [6]University of Geneva, Geneva, Switzerland; [7]Johns Hopkins University, Baltimore, USA

Conflict of Interest:
The author has declared no conflict of interest.

Hepatitis E

- Hepatitis E exists worldwide, large outbreaks in Africa and Asia
- Long incubation period, mean 5 – 6 weeks (range from 2-10 weeks)
- Usually self-limiting but can lead to acute liver failure and death
- Pregnant women in the second or third trimester, are at increased risk of acute liver failure, fetal loss and mortality
- No specific treatment

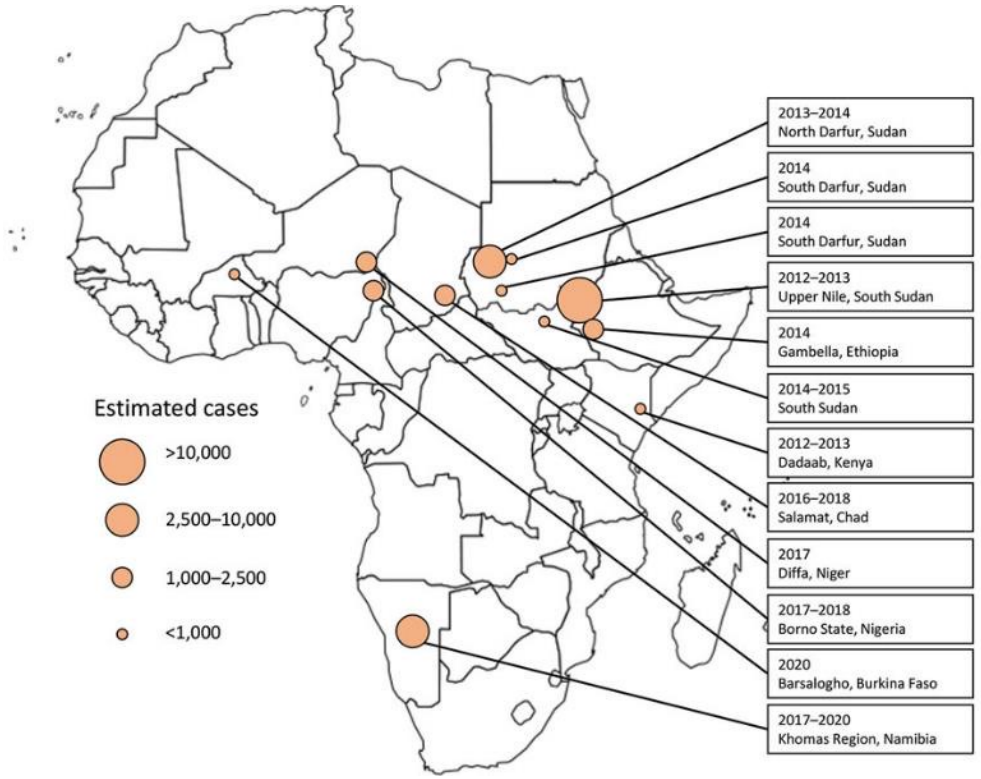


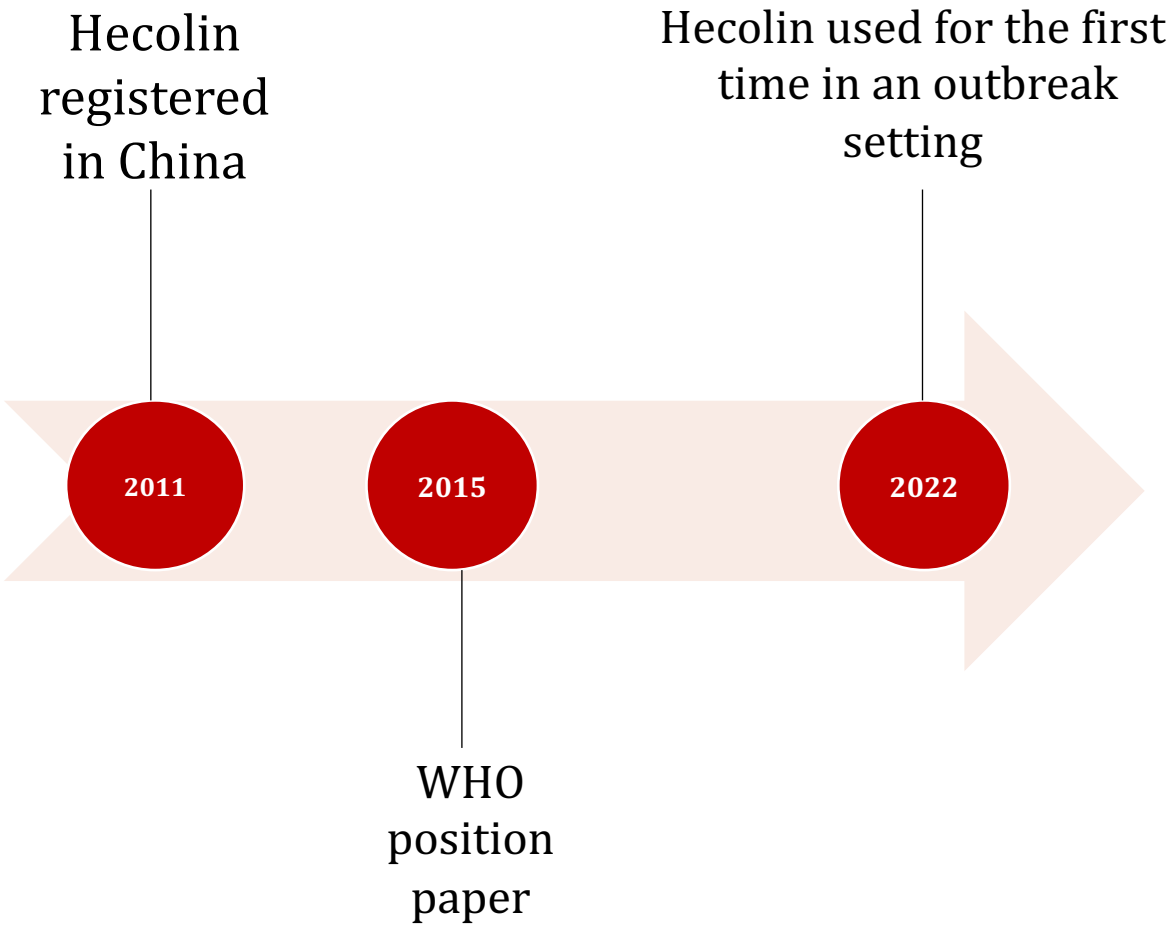
Figure. Geographic distribution of acute hepatitis E virus outbreaks reported among displaced persons in sub-Saharan Africa, 2010–2020
Source CDC Research Letter Viral Hepatitis E Outbreaks in Refugees and Internally Displaced Populations, sub-Saharan Africa, 2010–2020, May 2022

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Hepatitis E vaccine

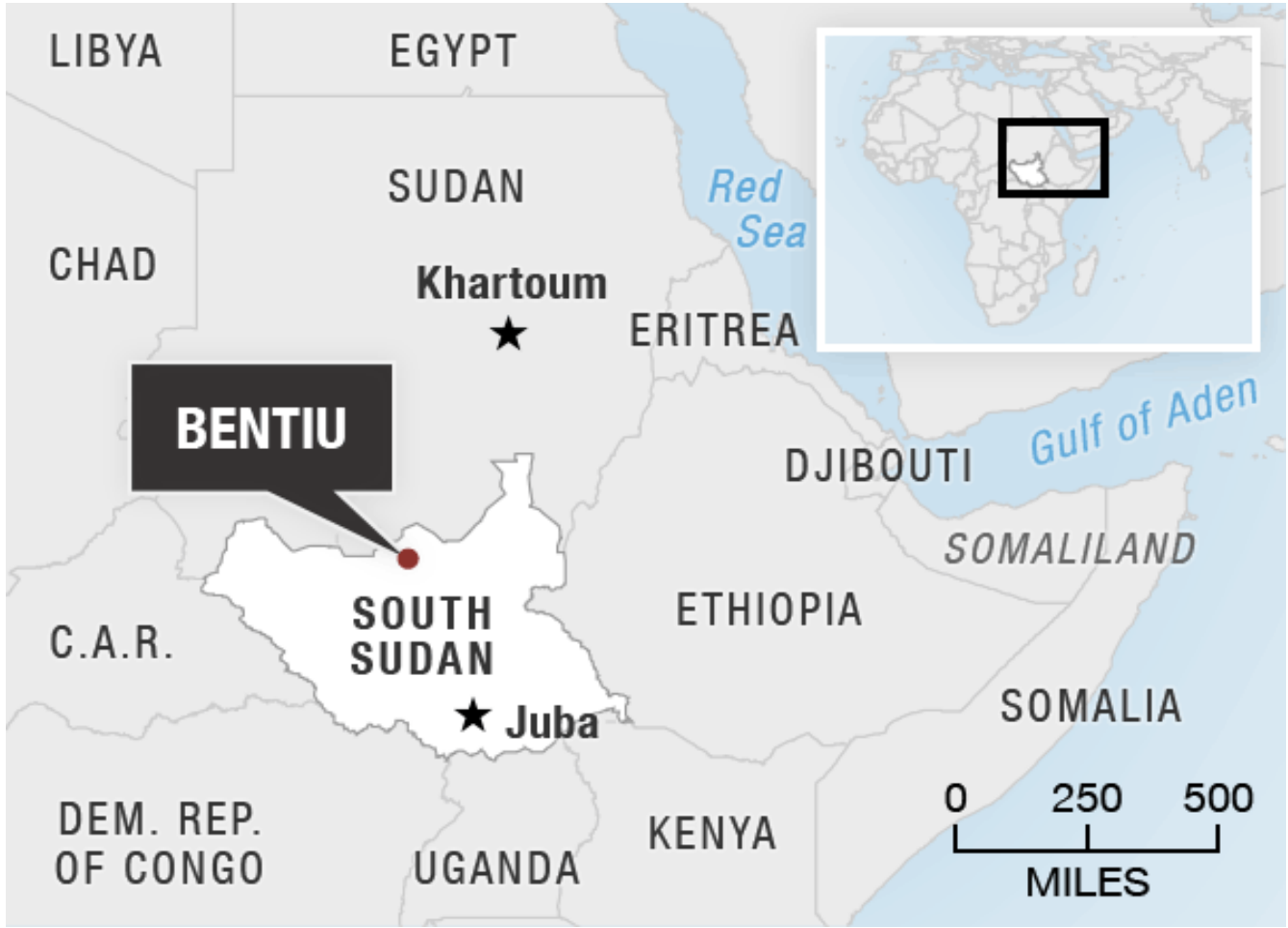
- Recombinant vaccine, Hecolin ® (Innovax, China)
- 3 doses (0, 1, 6 mo), ≥16 yrs old
- 100% efficacy in a per protocol analysis at 1 year and 93% at 54 months
- Not WHO prequalified
- WHO 2015 recommendation for use in outbreak setting



MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Bentiu internally displaced persons camp, South Sudan

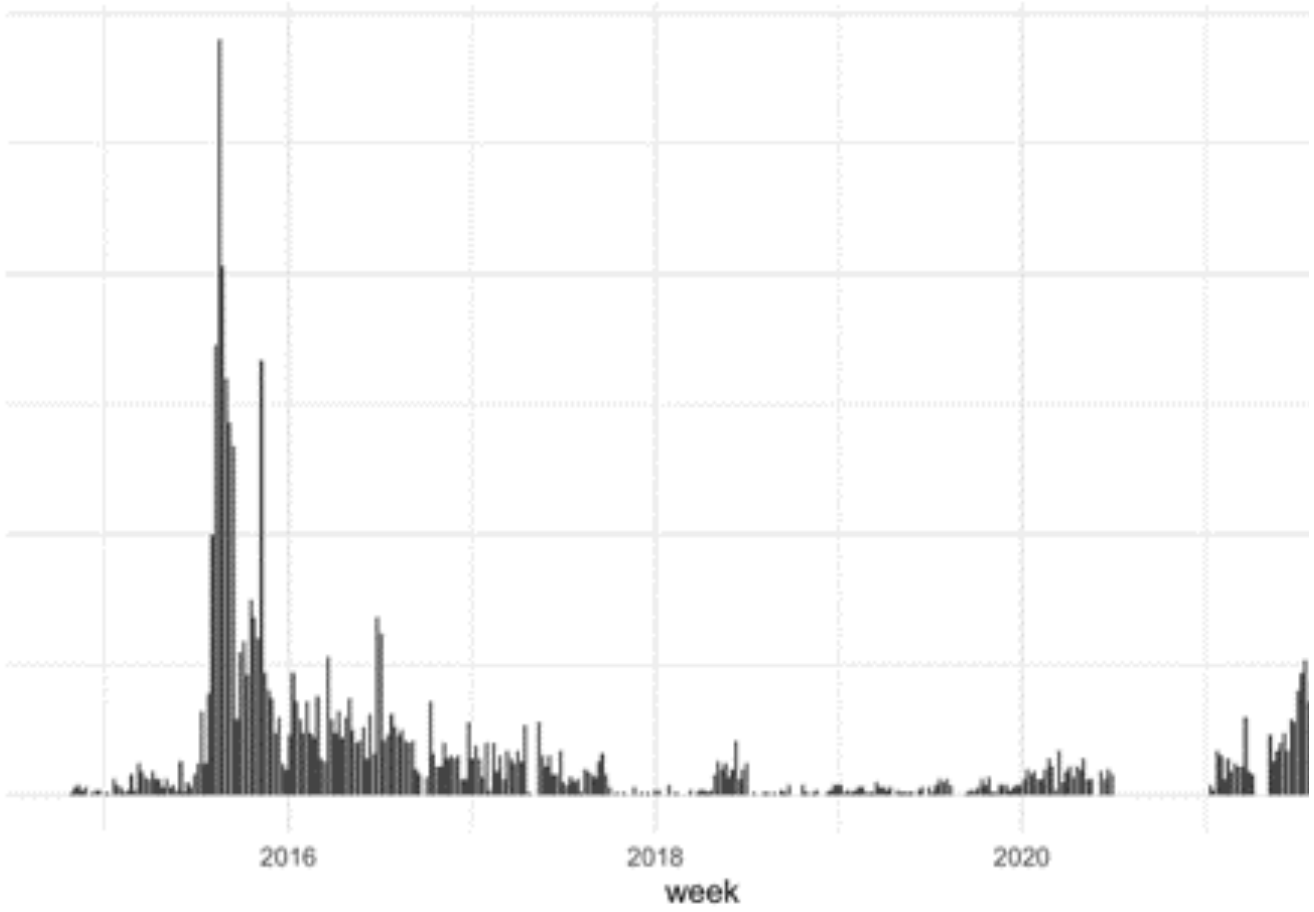


- Established Dec. 2013
- ~112,000 residents
- Limited water and sanitation facilities

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Hepatitis E in Bentiu internally displaced persons camp, South Sudan



- Large hepatitis E outbreak in 2015, with ongoing transmission after
- Increase of cases mid-2021 following floods and population influx
- Outbreak declared in August 2021
- MOH requested MSF to support integration of hepatitis E vaccination in outbreak control strategy

MEDECINS SANS FRONTIERES

First mass reactive vaccination campaign

Target population:

✓~ 27’000 Bentiu IDP camp residents

✓16-40 years old, including pregnant women

✓No acute illness or jaundice

- Implemented in 3 vaccination rounds: March, April and October 2022
- Vaccines delivered through fixed and mobile sites, and door-to-door strategy

➢ High administrative vaccination coverage >90% in each vaccination round

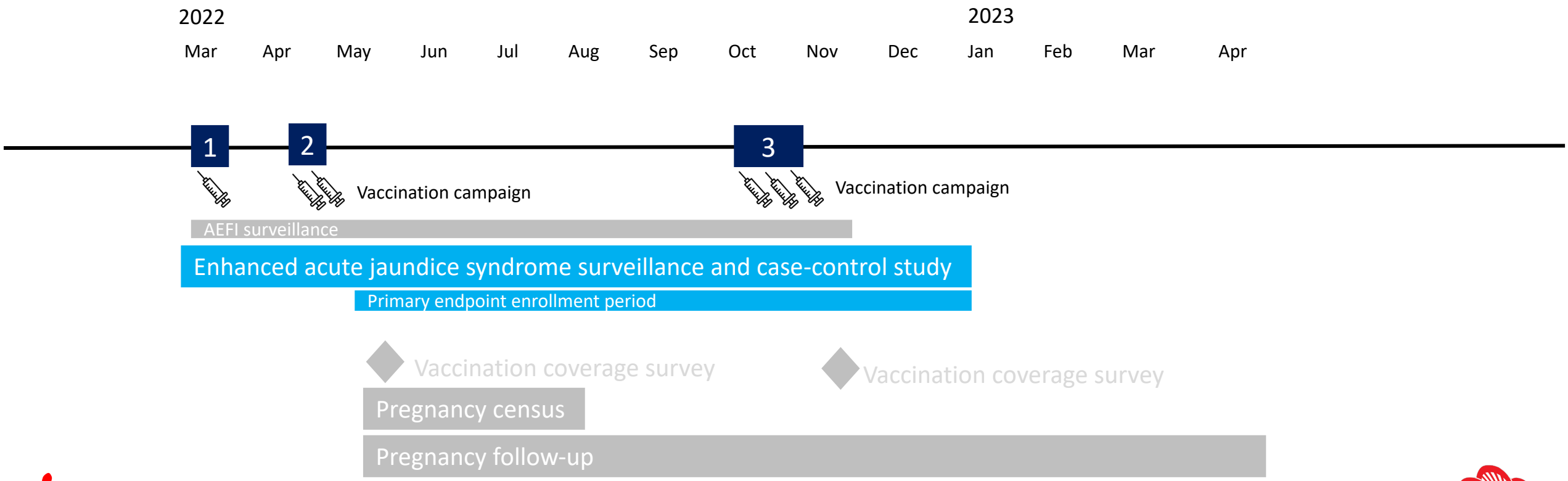
Objectives

1. Feasibility and acceptance
   - Vaccination coverage surveys
   - Focus group discussions
2. Safety
   - Adverse events monitoring during vaccination
   - Adverse events reported during survey
   - Pregnancy cohort to monitor pregnancy outcomes
3. Two-dose vaccine effectiveness
   - Enhanced surveillance acute jaundice syndrome
   - Case-control study

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Timeline of vaccination and study



Acceptance and safety

- High vaccination coverage
- Vaccine safe with few adverse events reported

Please see details in the poster

	According to recall or card		Confirmed by card	
Coverage by dose	**% (n)**	**95 % CI**	**% (n)**	**95 % CI**
One or more doses	86% (1377)	[84-88]	40% (644)	[37-43]
Two or more doses	73% (1160)	[70-75]	19% (305)	[17-21]
Three doses	58% (924)	[55-61]	10% (163)	[9-12]

Note that confirmed by card means that all doses reported were verified on vaccination card. Very few individuals retained vaccination cards or had them available at interview.

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Methods: Enhanced surveillance

All cases with acute jaundice seeking care at MSF hospital referred from clinician to study team

✓Written consent
✓Questionnaire
✓Vaccination status
✓Follow-up visit 2-4 weeks later

✓Lab testing in Bentiu
- Assure HEV IgM RDT
- Hepatitis C, B, malaria
- Liver function tests (**ALT**, *AST, Bilirubin*)

✓Lab testing in Geneva
- ELISA IgM
- ELISA IgG
- PCR

Suspected HEV case: An individual presenting with an acute (i.e., recent, new or sudden) onset of jaundice, dark urine or pale clay stools.

Probable HEV case: A suspected case with a serum alanine aminotransferase (ALT) concentration ≥2.5 times the normal range limit with

1) a positive IgM test (ELISA and/or RDT) test or;

2) a ≥4-fold rise in IgG titers (ELISA) in paired samples collected 2 to 4 weeks after initial blood draw.

Confirmed HEV case: A suspected or probable HEV case with PCR-detected HEV RNA isolated from a plasma sample.

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Methods: Case-control study

Vaccine eligible cases with acute jaundice seeking care at MSF hospital enrolled in surveillance

✓Written consent

✓Questionnaire

✓Vaccination status

✓Follow-up visit 2-4 weeks later

✓6 controls per case recruited in the community

✓Lab testing in Bentiu

- Assure HEV IgM RDT
- Hepatitis C, B, malaria
- Liver function tests (**ALT**, *AST, Bilirubin*)

✓Lab testing in Geneva

- ELISA IgM
- ELISA IgG
- PCR

Suspected HEV case: An individual presenting with an acute (i.e., recent, new or sudden) onset of jaundice, dark urine or pale clay stools.

Probable HEV case: A suspected case with a serum alanine aminotransferase (ALT) concentration ≥2.5 times the normal range limit with

1) a positive IgM test (ELISA and/or RDT) test or;

2) a ≥4-fold rise in IgG titers (ELISA) in paired samples collected 2 to 4 weeks after initial blood draw.

Confirmed HEV case: A suspected or probable HEV case with PCR-detected HEV RNA isolated from a plasma sample.

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Results: case enrolment

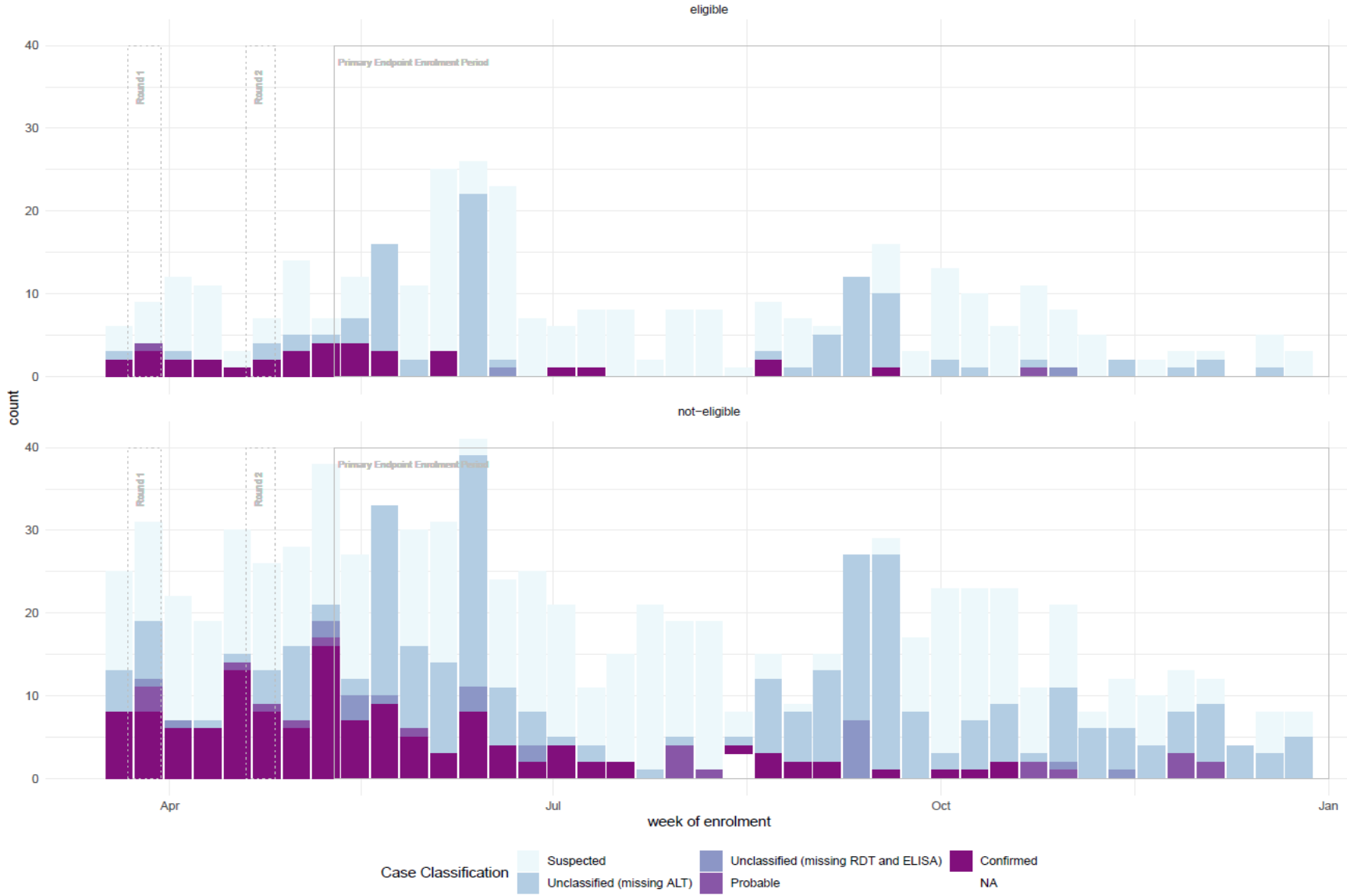
- Enhanced surveillance: all suspect cases seeking care at MSF hospital
- Case-control study : vaccine eligible suspect cases

Status	All suspect cases	Vaccine-eligible suspect cases
Enrolled	1,186	354
Admitted to hospital	137	26
Died	18	3

Results: case enrolment by vaccine eligibility

Vaccine-eligible cases
- Reside in Bentiu IDP camp
- Age 16 – 40 years

Non-eligible cases
- Children <16 years
- Adults >40 years
- Reside outside Bentiu IDP



MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Results: case enrolment by vaccine eligibility

17 (5.6%) out of 287 vaccine-eligible cases were probable or confirmed hepatitis E

	Eligible (N=287)	Not-eligible (N=625)
Sex		
Female	161 (56.1%)	252 (40.3%)
Male	126 (43.9%)	373 (59.7%)
Age Group		
0-5	0 (0%)	219 (35.0%)
6-10	0 (0%)	137 (21.9%)
11-15	0 (0%)	100 (16.0%)
16-39	277 (96.5%)	100 (16.0%)
40+	10 (3.5%)	69 (11.0%)
Effective Doses		
0	97 (33.8%)	612 (97.9%)
1	49 (17.1%)	3 (0.5%)
2	113 (39.4%)	3 (0.5%)
3	11 (3.8%)	0 (0%)
Missing	17 (5.9%)	7 (1.1%)
Study Classification		
Suspected	270 (94.0%)	542 (86.7%)
Probable	1 (0.3%)	14 (2.2%)
Confirmed	16 (5.6%)	66 (10.6%)
Missing	0 (0%)	3 (0.5%)

*Preliminary analysis results may change

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Results: preliminary* vaccine effectiveness estimate

Two doses of Hecolin vaccine provides 83.9% (-33.1–98.1) effectiveness against probable or confirmed cases

	Cases	Controls	Crude VE (95% CI)
1 Dose	15	48	87.9% (0.8–98.5)
2 Doses	13	44	83.9% (-33.1–98.1)
≥1 Dose	17	99	86.5% (36.3–97.1)

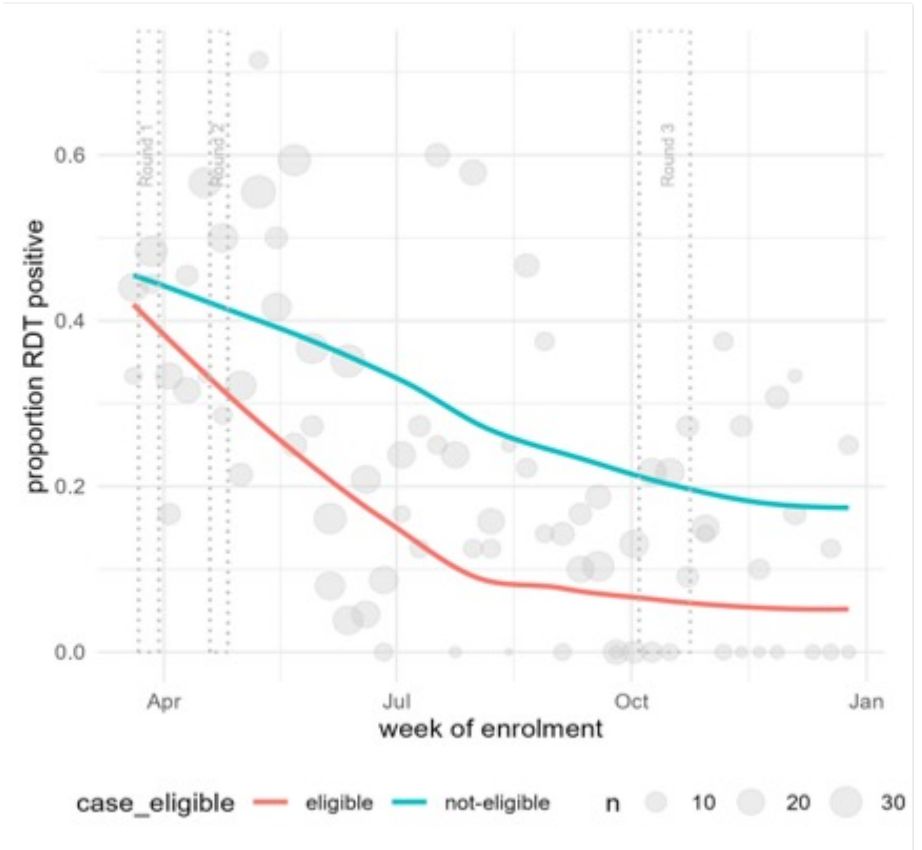
*Preliminary analysis results may change

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Results: reduction in incidence after vaccination

2.6-fold decrease in the incidence rate of RDT positive hepatitis E cases before and after the second dose campaign in Bentiu (including those not eligible for vaccination)



Discussion

Limitations

- Results are preliminary, analysis and confirmatory PCR testing is ongoing
- Low number of confirmed cases leads to large confidence intervals
- Specific Bentiu context
    - Population movement
    - Prior exposure to hepatitis E virus

Strengths

- Bias indicator analysis shows null vaccine effectiveness against non-hepatitis E jaundice
    - 2 dose VE: 9.8% (-34.6–39.5)
- Large amount of clinical data generated which will help us understand vaccine, diagnostics and general HEV epidemiology

MEDECINS SANS FRONTIERES

SCIENTIFIC DAYS

Conclusions

- Vaccine was well accepted and tolerated in Bentiu IDP camp community
- Preliminary estimates suggest short-term two-dose protection to be strong and potentially sufficient for outbreak response
- Reduction in case incidence overall after vaccination

➢Vaccination one of the interventions in outbreak response

Elizabeth Nyaruon Thon, the Chairperson of Bentiu IDP camp is exasperated over the sanitation conditions in the camp. © Peter Caton / MSF

Thank you to the South Sudan Ministry of Health, World Health Organisation, Health partners in Bentiu, MSF OCA and OCG in Bentiu, Juba, Amsterdam and Geneva

Thank you to the CHC and the entire population of Bentiu IDP camp

MSF staff member, Doki Simon, working on the Hepatitis E vaccination study. © Peter Caton / MSF

MSF SWITZERLAND HEPATITIS E VACCINATION TEAM, BENTIU

Gilles Grandclement , PC
Fati Oumarou, PMR
Jetske Duncker, PMR/DMedCo
Vinicius Ruas, HP
Dan Acheson, LM/DLogCo
Monika Egli, HR Fin Admin

MSF SWITZERLAND, BENTIU HOSPITAL STUDY TEAM

Vincent Kinya, Epidemiology Activity Manager
Doki Simon, Epidemology Activity Manager
Chuol Phar Met, Data processing Officer
Stephen Yoal Makon Machar, Clinial Research Assistant
Nyadiet Priscilla Luony, Clinical Research Assistant
Malual Nguen Chuol, Lab technician
Gatmai Reer,Survey team supervisor
Survey team

STUDY CO-INVESTIGATORS

John Rumunu Ministry of Health, South Sudan
Joseph F. Wamala World Health Organization, South Sudan
Frederick Beden Loro World Health Organization, South Sudan
Simon John Deng Chan World Health Organization, South Sudan
Mr. Peter Mahal Water Resources/ Rural Water Department, South Sudan
Robin Nesbitt, Epicentre
Iza Ciglenecki MSF Switzerland
Melat Heile MSF Switzerland
Patrick Nkemenang MSF South Sudan
Priscillah Gatimah, MSF South Sudan
Nicolas Peyraud MSF Switzerland
Manuel Albela MSF Switzerland
Monica Rull MSF Switzerland
Etienne Gignoux, Epicentre
Andrew Azman, MSF Switzerland, Johns Hopkins University
Aybüke Konyucu, Epicentre, Johns Hopkins University
Dr. Isabela Eckerle, Centre for Emerging Viral Diseases, Geneva University Hospitals
Catia Alvarez, Centre for Emerging Viral Diseases, Geneva University Hospitals

THANK YOU TO MSF OCA BENTIU HOSPITAL TEAM FOR THE COLLABORATION